

Ranjan Sarkar

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SUMMARY

PhD scholar at the Department of Artificial Intelligence (DoAI), IIT Kharagpur, in the **RHPI (Receding Horizon and Predictive Intelligence) Lab** under the supervision of **Dr. Prabhat Kumar Mishra**. My research focuses on developing new robust RL algorithms with applications in LLMs and robotics.

RESEARCH INTERESTS

Machine Learning, Optimization, Reinforcement Learning, State-Space Models

EDUCATION

PhD, Artificial Intelligence	2025 – Present
Indian Institute of Technology Kharagpur	CGPA: 9.33/10
MSc, Mathematics	2021 – 2023
Indian Institute of Technology Kharagpur	CGPA: 8.76/10
BSc, Mathematics	2018 – 2021
Ramakrishna Mission Vidyamandira (University of Calcutta)	CGPA: 9.15/10
Higher Secondary (Class XII), WBCHSE	2018
Raiganj Coronation High School (H.S.)	88.6%
Secondary (Class X), WBBSE	2016
Raiganj Coronation High School (H.S.)	94.57%

PUBLICATIONS

Conference Papers

[C1] Ranjan Sarkar, Prabhat K. Mishra. “LQG Solution for POMDP without estimating states: A minimum variance approach.” Accepted at *Indian Control Conference (ICC)*, 2027.

PROJECTS

Master’s Thesis: Image Compression	2022 – 2023
Supervisor: Prof. Swanand Khare, Department of Mathematics, IIT Kharagpur	
- Implemented tensor CP decomposition in Python. - Applied the Alternating Least Squares (ALS) optimization method to obtain low-rank approximations of RGB image tensors.	

WORK EXPERIENCE

Senior Business Analyst	2023 – 2024
EXL Service	Gurgaon, India
- Built data pipelines and analytics with SQL and SAS. - Worked on business-as-usual (BAU) operations.	

HONORS AND AWARDS

- Qualified Joint CSIR-UGC NET (June 2024) with All India Rank 83.
- INSPIRE Scholarship (2018–2023), DST, Govt. of India, awarded for top 1% in the WB HS board examination (2018).

TEACHING EXPERIENCE

Teaching Assistant, Department of Artificial Intelligence, IIT Kharagpur

Reinforcement Learning	Autumn 2026
AI for Robot Autonomy	Autumn 2026
AI for Cyber Physical Systems	Spring 2026
Algorithmic and Mathematical Foundations for AI	Autumn 2025

ACADEMIC SERVICE

Event Organization

Organizing Committee Member and Volunteer , Departmental Foundation Day and Research Scholar Day 2026, Department of Artificial Intelligence, IIT Kharagpur	Aug 2026
Volunteer , Workshop on AI for Science and Technology 2026, IIT Kharagpur, by IndiaAI	Jan 2026

TECHNICAL SKILLS

Programming Languages: Python, C, SQL, SAS

ML and Scientific Libraries: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, SymPy, CasADi, Gymnasium

Tools: Jupyter, LaTeX

RELEVANT COURSEWORK

Machine Learning, Deep Learning, Artificial Intelligence, Convex Optimization, Non-Linear Programming, Linear Algebra, Probability and Statistics, Real Analysis, Advanced Numerical Techniques, AI for Cyber Physical Systems, AI for Robot Autonomy

LANGUAGES

Bengali, English, Hindi

INTERESTS

Photography, Singing, Swimming, Table Tennis